

AVIAN REPRODUCTIVE DISEASES

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1. ANATOMY OF THE REPRODUCTIVE SYSTEM

The female reproductive tract is composed of the left ovary and oviduct. The left ovary is located at the level of the cranial division of the kidney.¹ It is suspended from the dorsal body wall by the mesovarian ligament. The immature ovary is flat and has a smooth texture. As the ovary matures the developing follicles give the ovary its typical grape-cluster like appearance. During the non-breeding season the ovary diminishes in size considerably. The ovary receives arterial blood from the ovario-oviductal branch of the left cranial renal artery. It is drained by two ovarian veins that empty into the caudal vena cava. The blood vessels enter the ovary at the ovarian hilus, which is located on the dorsal surface of the ovary. The oviduct is made up of five individual regions: the infundibulum, the magnum, the isthmus, the uterus, and the vagina. The oviduct occupies most of the left dorso caudal part of the coelomic cavity. Birds with seasonal breeding cycles show variation in the size of the oviduct between the breeding and non-breeding season. Ovarian hormones, estrogen, progesterone, and androgens control the seasonal change in oviductal size.² The oviduct is suspended from the dorsal wall of the coelomic cavity by a double-layered sheet of peritoneum forming the dorsal and ventral ligaments of the oviduct. The dorsal ligament extends from the dorsal part of the coelomic cavity to the oviduct. In immature birds of many species the dorsal ligament can be seen crossing over the cranial division of the kidney. The identification of the dorsal ligament is useful when endoscopically sexing reproductively immature birds. The ventral ligament outstretches ventrally from the oviduct and has a free border. Arterial blood reaches the oviduct through the cranial, middle, and caudal oviductal arteries.¹ The cranial artery is a branch of the left cranial renal artery, the middle artery is a branch of the left ischiatic artery, and the caudal artery is a branch of the left pudendal artery. The oviductal arteries form longitudinal anastomosing arterial arcades along the dorsal and ventral surfaces of the oviduct in the dorsal and ventral ligaments. The uterus is the most vascular section of the oviduct. The oviductal veins draining the cranial part of the oviduct empty into the caudal vena cava, while the ones draining the caudal section empty into the hepatic portal and renal portal systems.¹

The male reproductive tract is composed of paired testicles, epididymis, and ductus deferens and in some species a phallus. The testicles are bean-shaped and located in the dorsal coelomic cavity, on either side of the midline and ventral to the cranial division of the kidneys. Each testicle is suspended by a short mesorchium. The testicles are located within the intestinal peritoneal cavity and partially surrounded by the abdominal air sac. The size of the testicles increases dramatically with sexual activity. This enlargement is due to an increase in length and diameter of the seminiferous tubules and to a greater number of interstitial cells.³ The color of the testicles may vary with sexual activity. The immature testicle may be yellow from the accumulation of lipids in the interstitial cells.³ Melanistic testes, like melanistic ovaries, can occur in some species of Psittaciformes (golden conure, blue and gold macaw, some cockatoos),

Passeriformes and Piciformes (keel-billed toucans).⁴ As the testicles enlarge during sexual activity the color may change due to the scattering of the interstitial cells and/or melanocytes by the growing seminiferous tubules. The surface of the testicle is covered by the tunica albuginea. Unlike mammals, birds do not have testicular lobulation.¹ Blood is supplied to the testicles by the testicular artery, a branch of the cranial renal artery. Birds do not have a pampiniform plexus. Blood from the testicles drains into the caudal vena cava through several short testicular veins. During the breeding season the blood supply to the testicles increases dramatically causing a prominent vascular pattern on the testicular surface. Some avian species have a copulatory organ, the phallus. There are two types of phallus in birds, intromittent and non-intromittent.³ The intromittent type occurs in ratites, tinamous, anseriforms, black buffalo weavers, curassows, and guans. The non-intromittent type occurs in some domestic galliformes and some passerine birds. The phallus is located on the ventral surface of the proctodeum. Psittaciformes do not have a phallus. Instead, copulation is achieved through eversion of the cloacal wall and transfer of sperm from the ejaculatory papilla to the female cloaca.

2. REPRODUCTIVE DISORDERS IN THE FEMALE

2.1 Chronic egg laying and dystocia

This condition occurs when a hen lays an excessively large number of eggs over an extended period, despite the absence of a mate or breeding season. Chronic egg laying is most commonly seen in smaller species such as cockatiels, lovebirds, budgerigars, and finches. If not treated this problem will lead to serious consequences including, malnutrition (particularly calcium deficiency), dystocia, egg yolk peritonitis, and osteoporosis.

The etiology is multifactorial. Many times environmental stimulation such as increased photoperiod and the presence of a perceived mate or nesting site can lead to the activation of the reproductive hormones resulting in the production of eggs. It has also been suggested that a genetic predisposition and/or abnormal reproductive hormone balance can lead to chronic egg laying.

Early clinical signs include increased frequency in between clutches, increased number of eggs per clutch, and increased breeding behavior and broodiness. Unfortunately many birds are presented at an advanced stage. Clinical signs of chronic egg laying include, in addition to the early signs, weight loss, dehydration, depression, lethargy, weakness, pathological fractures, or evidence of egg binding and/or dystocia.

The treatment of chronic egg laying must include proper environmental modification in addition to pharmaceutical and/or surgical treatment. Providing an adequate diet is also key for a successful outcome. There have been a variety of pharmaceuticals used to treat chronic egg laying with various degrees of success. The use of medroxyprogesterone is no longer advocated due to the severe side effects observed, which include polyuria, polydipsia, obesity, lethargy, hepatic lipidosis, hepatic cirrhosis, diabetes mellitus, and death.¹² Human chorionic gonadotropin (HCG; Pregnyl, St. Louis, MO, USA) has minimal side effects, however, the results are less consistent and some birds may become refractory to repeated treatments. Leuprolide acetate is a long-acting gonadotropin-releasing hormone analog that has proved effective in controlling excessive egg laying without significant deleterious side effects. The results are sometimes inconsistent, with some birds needing longer therapy than others. Tamoxifen, an estrogen blocker used to treat women with breast cancer, was evaluated as a potential treatment in budgerigars for

chronic egg laying.¹⁴ The most significant side effect was leukopenia that resolved when the treatment was discontinued.

Leuprolide acetate is the pharmaceutical used at our clinic to treat chronic egg laying. I administer leuprolide (700-1000 µg/kg, IM) every 14 days for 3 treatments. Most birds respond to this protocol and stop egg laying for the rest of the season. Some birds require regular injection intervals for a longer period. The environmental modification must be part of any successful treatment plan. Shortening the light cycle to 8-10 hours and eliminating the source of sexual stimulation may help decrease reproductive hormones activity. Sexual stimulation can occur by excessive petting by owner, and/or the presence of a nest or toy that may resemble a nest.

Deslorelin acetate implants are indexed in the US for ferrets for adrenal endocrinopathy. There are a number of studies performed in poultry, quail, and pigeons and psittaciformes. Implantation of a 4.7 mg deslorelin implant in chickens (*Gallus gallus domesticus*) resulted in 100% cessation of egg laying by 14 days with suppressed ovarian activity of 180 days (range, 125–237 days).⁵ In Japanese quail (*Coturnix japonica*), 4.7-mg was less consistent approximately 60% to 70% efficacious and only lasted 2 to 4 months.^{6,7} Few clinical reports and retrospective studies in psittacine species exist. In one large retrospective study study, 32 Psittaciformes were administered a 4.7-mg deslorelin implant that was 100% efficacious and suppressed egg laying for up to 3 months.⁸ In a study n cockatiels (*Nymphicus hollandicus*) a 4.7-mg deslorelin acetate implant suppressed egg laying in healthy for at least 180 days.⁹ Similar results were found in Quaker parrots (*Myiopsitta monachus*), receiving a 4.7 mg deslorelin implant and suppressing egg laying for at least 6 months.¹⁰

Salpingohysterectomy is recommended in refractory cases and must also be combined with environmental modification and hormonal treatment. Removing the oviduct does not cause the cessation of follicular maturation within the ovary.

2.2 Oophoritis and salpingitis

Oophoritis or ovarian inflammation is often caused by bacterial agents, and less commonly by viral, and rarely fungal or parasitic agents. Oophoritis usually results from spreading infections from the peritoneal cavity or air sacs or can result from hematogenous spread. Affected ovaries are usually enlarged, wrinkled, discolored, and hemorrhagic. Inspissated yolk can sometimes be found within the ovarian follicles. Noninfectious oophoritis has been associated with follicular rupture and spilling of yolk material into the ovarian stroma. The affected ovaries are usually enlarged with multifocal yellow foci.

The clinical include depression, anorexia, weight loss, abdominal distention, dyspnea, and sudden death. The diagnosis of oophoritis can be challenging. Radiographs, followed by endoscopic examination are the most reliable diagnostic tools (which could be contraindicated with fluid in the coelomic cavity). Antibiotics, anti-inflammatories and supportive care (fluids, nutritional support, and warmth) should be administered as indicated. GnRH agonists has been suggested as an adjunctive treatment as well.¹¹

Salpingitis or inflammation of the oviduct can involve the whole organ, or in some cases it may be localized to the uterus (metritis). Viral, bacterial, fungal, and parasitic infections can affect the oviduct. The organisms can originate at the cloaca, air sacs, peritoneal cavity, or blood. Noninfectious causes include trauma, improper egg formation, and inspissated yolk. Affected oviducts are usually enlarged, reddened, and friable. Salpingitis and/or metritis are common causes of dystocia or can result as sequela to egg binding and dystocia.

Clinical signs include depression, weight loss, abdominal distention, tenesmus, infertility, and abnormally shaped and/or colored eggs. Hematology, radiography, and ultrasonography are useful diagnostic tools. However, early diagnosis can be challenging and, as a result, many birds are presented when the disease is at an advanced state. Multiple organ systems are often affected at this stage due to oviductal rupture and/or septicemia. Treatment is based in antibiotics, anti-inflammatories and supportive care. In severe cases salpingohysterectomy may be required.

2.3 Ovarian cysts and cystic oviduct

Ovarian cysts are seen in a number of species including cockatiels, budgerigars, canaries and ducks.¹² The exact etiology is unknown. Endocrine disorders, degenerative disorders and neoplasia can lead to cyst formation. Clinical signs can include polyuria/polydipsia, coelomic distention, dyspnea, and decreased or cessation of egg production. Cystic ovaries are often seen as an incidental finding during endoscopic examination. Radiographs may show polyostotic hyperostosis and ventrocaudal displacement of the abdominal viscera (if large cysts are present). Ultrasonography is helpful in differentiating coelomic distention due to fluid, mass, or an egg. Large cysts can be surgically resected, or the cyst content can be aspirated. GnRH agonists has been suggested as an adjunctive treatment as well.¹¹

2.4 Ovarian and oviductal neoplasia

There are several tumors that can affect the ovary. The most commonly reported type is granulosa cell tumor.¹³ They usually appear as friable, yellow, and lobulated irregular masses. The tumors can become very large and fill the abdominal cavity. Ovarian carcinomas are variably sized, firm, gray-white, and multilobular. They may spread locally to adjacent organs, the mesentery, or the body wall, and can also metastasize. Ovarian teratomas can be large and are comprised of various tissue types. Hemangiomas and leiomyomas of the mesovarium have also been reported.

Clinical signs include coelomic distention, dyspnea, and ascites. The diagnosis is often based on ultrasonographic or radiographic findings and endoscopic examination. Cytologic evaluation of the ascites may show the presence of neoplastic cells. Ovarian tumors carry a poor to grave prognosis. GnRH agonists has been suggested as an adjunctive treatment as well.¹⁴

Adenomas and carcinomas are commonly reported in the oviduct.^{13,15} They are usually firm, nodular, grey-white masses. These tumors can spread to the peritoneal surfaces. Grossly it is difficult to differentiate tumors of ovarian versus oviduct origin. Leiomyomas have been reported in the oviduct of chickens.¹³

Clinical signs include abdominal distention, dyspnea, ascites, and weight loss. Ultrasonography and radiography can provide useful information as to the nature of the enlargement. Ultrasonography is particularly useful in the differentiation of solid masses versus a cyst or abscess. Endoscopic examination and/or exploratory laparotomy will help identify the precise location of the tumor. Salpingohysterectomy is the best treatment option. If the tumor is confined to the oviduct the prognosis is good. Sometimes the tumors may involve the ovary, in which case the prognosis is guarded to poor.

2.5 Dystocia

Dystocia could be obstructive or non-obstructive (sometimes referred as egg binding). It usually involves the uterus, the vagina, or the opening of the vagina into the cloaca. Egg binding can result from malnutrition, obesity, stress, and systemic disease. Dystocia is often associated

with large and/or malformed eggs, damage and scarring to the oviduct from previous problems, uterine torsion, and/or salpingitis.¹³ Dystocia often constitutes an emergency as the lodged egg can cause excessive pressure on the pelvic blood vessels, disrupt renal circulation, obstruct the intestinal tract, and can lead to shock.

Cockatiels, budgerigars, lovebirds, canaries and finches are most commonly affected. These conditions are more serious in smaller birds probably because of the faster onset of metabolic compromise.

Clinical signs include depression, fluffed feathers, wide stance, dyspnea, lameness, paresis or paralysis, and tenesmus. The diagnosis is based on history, clinical signs, physical exam, radiographs, and ultrasonography. Polyostotic hyperostosis has been described in patients with ovarian tumors and is thought to be associated with secreting large amounts of estrogens;¹⁶ however, the association has been refuted in another publication.¹⁷ Also these diagnostic tools can help identify the presence of multiple eggs in the oviduct, which is not that uncommon. Often, an egg may be palpated within the coelomic cavity. However, if the egg is soft-shelled or located in the cranial oviduct it may not be palpable.

The treatment will depend on the physical condition of the patient and the severity of the clinical signs. Initial treatment should be aimed at stabilizing the patient. Supportive care including fluid therapy, thermal and nutritional support might be needed. For non-obstructive dystocia, parenteral calcium can be administered SQ with the fluids. The cloaca is lubricated with a water-based lubricant. In many cases of egg binding initial supportive care is enough to hasten the passage of the egg through the oviduct. If the bird is stable, but the egg remains in the oviduct, 3 therapies that have been described: oxytocin, prostaglandin E2 (PGE2), and prostaglandin F2a (PGF2a). La oxitocina is most often used, but does not relax the uterovaginal sphincter; therefore, it is used only if the sphincter is open, like PGF2a. Prostaglandin E2 (PGE2, Prepidil gel, Upjohn Co, Kalamazoo, MI, USA) can be applied topically to the cloacal mucosa to induce relaxation of the uterovaginal sphincter and uterine contractions leading to oviposition. This product is used to induce labor in women and should be handled with care because of the potential deleterious effect on pregnant women. In some cases, under sedation or general anesthesia, the egg can be gently manipulated caudally if it can be visualized through the cloaca. For obstructive dystocia, ovocentesis may be required. To perform ovocentesis, a large gauge needle (20-18 gauge) is inserted and the content of the egg is aspirated while applying gentle finger pressure to collapse the shell. The shell fragments are usually passed within a few days. If the egg is not visible through the vent, ovocentesis can be performed transcutaneously but could result in significant oviductal pathology. Severe cases of dystocia may need salpingotomy or salpingohysterectomy.

2.6 Yolk peritonitis

Egg yolk peritonitis occurs when yolk material is accidentally deposited within the intestinal peritoneal cavity. Many factors can lead to this condition. Salpingitis can lead to reverse peristalsis and spilling of the oviduct content through the funnel of the infundibulum and into the intestinal peritoneal cavity. Rupture of the oviduct can also result in leakage of egg material into the peritoneal cavity. Trauma can lead to ovarian follicle rupture and leakage of yolk into the coelom. Egg yolk peritonitis can be aseptic, but in most cases bacteria quickly colonize the rich yolk resulting in septic peritonitis.

Clinical signs include depression, lethargy, loss of body condition, abdominal distention, and dyspnea. The diagnosis is based on history, physical exam findings, clinical signs, and the fluid evaluation (including cytology) of the coelomic exudate. Radiographs may be useful, but

the presence of ascites often obscures serosal detail. In these cases, ultrasonography is best for assessing the coelomic cavity and the reproductive tract.

If the bird is severely dyspneic, aspiration of the ascites will improve the dyspnea. Supportive care with fluid therapy, thermal and nutritional support might be needed. Antibiotics might be indicated to treat underlying infection or prevent it in severe cases. In many cases multiple coelomocentesis are needed for management of the case. Refractory cases or cases in which a large amount of inspissated yolk is present within the peritoneal cavity may need coeliotomy to lavage the intestinal peritoneal cavity with care of damaging the integrity of the air sac walls to avoid fatal complications from fluid spills into the lungs.

2.7 Oviductal prolapse

Prolapse of oviductal tissues can occur as a complication to chronic egg laying, oviductal inertia, egg binding and dystocia, and space-occupying masses. Prolapses often occur secondary to excessive or prolonged straining from undesired behaviors such as self-stimulation and from medical causes. Treatment is aimed at stabilizing the patient with fluid therapy, thermal and nutritional support, and pain management. The prolapsed tissues must be protected and cleansed with warm saline and/or hypertonic solutions. If the prolapsed tissues are viable, should be gently replaced as soon as possible. If the tissues cannot be immediately replaced, they should be kept lubricated such as with a sterile surgical lubricant gel until surgery. Sutures across the vent may be necessary to keep the tissues in place after reduction, while allowing enough space between the sutures to defecate. Bloodwork and survey radiographs are recommended to evaluate underlying causes. Antibiotics and anti-inflammatories should be administered. Recommendations useful for the management of chronic egg laying often apply to cases of oviductal prolapse.

3 REPRODUCTIVE DISORDERS IN THE MALE

3.1 Orchitis

Inflammation of the testicles (orchitis) can be due to non-infectious and infectious diseases. Bacterial infections can result from extension of infections from the air sacs or peritoneal cavity, or hematogenous spread.¹³ Ascending infections from the phallus or cloaca can also result in primary orchitis. Affected testicles are usually enlarged, reddened, and may contain multifocal yellow-white foci.

The clinical signs are usually nonspecific and many birds may only show clinical signs if the disease is systemic. In breeding birds the most common clinical finding is decreased fertility. The diagnosis of orchitis is based on endoscopic examination and testicular biopsy collection for histopathologic evaluation and culture and susceptibility testing. Antibiotics and anti-inflammatories might be required.

3.2 Neoplasia

Testicular neoplasia is a common finding in companion psittacine birds. There are several testicular tumors that have been reported. These include seminoma, sertoli cell tumor, interstitial cell tumor, undifferentiated sarcoma, and teratoma.¹³ The highest incidence of testicular tumors is found in the budgerigar, but many other birds can be affected.

Clinical signs associated with testicular tumors include paresis or paralysis, abdominal enlargement, and dyspnea. Clinical signs of feminization have been reported in birds with Sertoli

cell tumors and include change in cere color in budgerigar and polyostotic endosteal hyperplasia.⁴ Interstitial cell tumors are often yellow-red due to steroid hormone precursors within. The diagnosis of testicular neoplasia is based on radiography, ultrasonography, and ultimately endoscopic examination and biopsy for histopathologic evaluation. Medullary bone formation in pneumatic and nonpneumatic bones can occur in male budgerigars with testicular neoplasms. Radiographic increases in medullary bone density, particularly in the humerus, could provide antemortem indication of testicular neoplasia in male budgerigars.¹⁸ Orchidectomy is the treatment of choice. Large tumors carry a poor to grave prognosis.

3.3 Phallus prolapse

Phallus prolapse is most commonly seen in ratites and anseriform birds. The prolapse is usually the result of trauma to the phallus and secondary bacterial infections. Extremely low environmental temperature has also been associated with prolapse and secondary frostbite. The prolapsed phallus should be thoroughly cleaned and disinfected. Local and systemic antibiotic therapy should be instituted. The phallus can be placed in the cloaca and held in place with a purse string suture. If severe trauma and/or necrosis are present amputation may be the best treatment option.

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AVIAN URINARY DISEASES

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1. ANATOMY OF THE URINARY SYSTEM

The paired kidneys are located dorsally in a depression of the synsacrum. Each kidney is made up of three divisions (cranial, middle and caudal). The divisions are composed of lobules with a large cortical mass and a small medullary mass. It is difficult to demarcate between the cortical and medullary portions of the lobules. The cortex is composed of both reptilian-type nephrons that do not contain loops of Henle and mammalian-type nephrons that do contain loops of Henle. The reptile-type nephrons are most numerous, and birds are less efficient in the excretion of electrolytes than mammals. The cortical-type nephrons are uricotelic and the medullary-type produce urine. The former are located on the surface of the kidney, and the latter are located in a deeper orientation. The nephrons' collecting ducts and ureters are contiguous. The kidneys of birds are larger by weight than in mammals. There are three pairs of renal arteries. The anterior branch arises from the aorta, and the middle and posterior branches arise from the sciatic artery or external iliac artery. The anterior branch supplies the cranial division of the kidney, and the middle and posterior branches supply the middle and caudal divisions of the kidney.¹

Birds have a renal portal system in which the renal portal vein functions like an artery by carrying blood to the tubules. Flow of blood into the kidneys from the renal portal system is controlled by valves. These valves are bilateral and are located at the junction of the external iliac vein and the renal vein. Studies suggest that acetylcholine causes the valves to close and epinephrine causes them to open. If agents excreted by the renal tubules are injected into the legs, they could be excreted by the tubules on the injection side of the body before entering the general circulation.¹

The avian kidney has a reduced capacity to secrete creatinine in comparison to uric acid. Renal output varies with the water intake and stress levels of the bird, but is generally considered to be 100 to 200 ml/kg/day. Some urine water that is excreted into the cloaca is passed by antiperistaltic movement of the cloaca into the colon where absorption of the liquid occurs.¹

2. ETIOLOGY AND PATHOPHYSIOLOGY OF URINARY DISORDERS

Renal disease in birds is an under-recognized problem. There are a number of different etiologies associated:

Infectious

Bacterial infections are one of the most common causes of nephritis in bird and can occur from a systemic infection. It can also occur from ascending infection since the ureters empty in

the cloaca or from gastrointestinal infection because the renal portal system receives afferent blood supply from the caudal mesenteric vein. A wide range of Gram-positive and Gram-negative bacteria is known to cause kidney disease. *Staphylococcus* spp., *Streptococcus* spp., *Listeria* spp., *Escherichia coli*, *Klebsiella* spp., *Salmonella* spp., *Yersinia* spp., *Proteus* spp., *Citrobacter* spp., *Edwardsiella* spp., *Enterobacter* spp., *Morganella* spp., *Providencia* spp., *Serratia* spp., *Pasteurella* spp., *Pseudomonas* spp., *Mycobacterium* spp., and *Chlamydia psittaci*.¹

Viral infections are usually multisystemic, although some viruses like avian polyomavirus, adenovirus and infectious bronchitis virus in chickens demonstrate a tropism for the kidney. Other viruses that have been associated with renal lesions include West Nile virus, Newcastle disease virus, paramyxovirus of pigeons, reoviruses, viruses belonging to the leukosis/sarcoma group and herpesviruses.

Fungal infections affecting the kidneys are commonly associated with an extension of a fungal airsacculitis or systemic infection, however, the kidney is rarely involved in either disease condition.¹ *Aspergillus* spp. are the most frequently implicated fungus in these lesions, but *Penicillium chrysogenum* and *Cryptococcus neoformans* have also been identified in kidney lesions.² Microsporidia, most commonly *Encephalitozoon hellem*, has also been reported in a number of avian species with variable effects in the kidney.

Parasitic nephritis due to *Eimeria*, *Cryptosporidium* spp. has been reported in a variety of avian species. *Eimeria truncata* is a well-known cause of renal coccidiosis in geese. Adult trematodes of *Tanaisia bragai* and *Paratanaisia bragai* have been found in collecting ducts in a number of avian species.

Inflammatory

Renal amyloidosis is most frequently observed in waterfowl and small passerine birds. Multiple organs, in addition to the kidney, are generally involved. Grossly there may be no discernible change, but, in severe cases, the kidneys are enlarged, pale, and somewhat friable.²

Toxicity

Renal disease may occur as a secondary condition due to heavy metal toxicity. Zinc-induced renal disease is nonspecific but may include the loss of renal architecture, apoptosis of individual cells, and acute tubular necrosis. Lead-induced renal lesions can

include proximal tubular necrosis and degeneration, visceral gout, and acid-fast intranuclear inclusion bodies. Mercury has been associated with necrosis of proximal tubular cells in birds

Mycotoxins such as aflatoxin B1 (*A. flavus* and *A. parasiticus*), ochratoxin A (*A. ochraceus* and *Penicillium viridicatum*), oosporein (*Chaetomium trilaterale*) and citrinin (*P. citrinum*) can cause multiple organ damage, including renal disease.

Drug-induced renal disease is primarily related to the use of NSAIDs and antibiotic medications. Among antibiotic agents, aminoglycosides may be the most problematic. NSAIDs like diclofenac, ketoprofen, flunixin meglumine have been associated with renal disease in birds, while others like meloxicam have had a low incidence of renal adverse effects.

Anomalous

Polycystic kidneys have been reported in birds but are usually an incidental finding.

Metabolic

Dehydration is an important contributor to renal disease. Severe or persistent dehydration increases resorption of water causing a subsequent reduction in urine flow. As uric acid secretion decreases, urates may precipitate in renal tubules and ureters leading to impaction and potentially renal failure.

Renal lipidosis has been correlated with high-fat diets and chronic liver disease. The latter is relatively common in Amazon parrots and cockatiels that have a high incidence of chronic-active hepatitis. Grossly the kidneys are pale, and microscopically there is fat in tubular epithelial cells.²

Neoplastic

Although kidney neoplasms have been reported in several bird species, renal tumors in budgerigars constitute a remarkably high proportion of all tumors in this species. Renal carcinoma is the most common tumor of the kidney; however, adenoma, nephroblastoma, cystadenoma, fibrosarcoma, lymphoma, and other neoplasms have also been reported in avian species

Nutritional

Vitamin A deficiency causes metaplastic changes of the ureters and collecting ducts as well as decreased secretions of mucus within the ureter that can ultimately lead to partial or complete ureteral obstruction and secondary hydronephrosis, hyperuricemia, and oliguric/anuric renal failure. Hypervitaminosis due to the excessive supplementation of vitamin D in the diet or from exposure to cholecalciferol rodenticide baits causes an excessive calcium uptake from the intestines, mimicking hypercalcinosis. High calcium intake also has been associated with renal disease, and anecdotal evidence that some species like budgerigars might be more susceptible.

Degenerative

Nephrosis, a non-specific histopathologic change characterized as any degenerative non-inflammatory lesions in the kidney, is a common finding in birds and might be associated with some causes mentioned previously but not necessarily.

Resulting from renal disease, gout has two forms in the avian species: visceral and articular gout. Visceral gout occurs as a result of increased plasma uric acid levels, resulting in deposition of urates on various organs, particularly the pericardium, liver, spleen, and kidney. Visceral gout may appear as a white coating when on the capsular surface of affected tissue. Articular gout occurs when uric acid crystals accumulate within the synovial capsules and tendon sheaths of the joints. In cases of articular gout, deposition of urates in the viscera typically does not occur. However, visceral and articular gout can be present in the same bird. Clinical signs include lameness, inability to ambulate, and swellings along the metatarsophalangeal and interphalangeal joints. Spontaneous articular gout in birds without underlying renal pathology is relatively uncommon and appears to have a hereditary basis, at least in chickens.³

Urolithiasis, as the formation of large urate “stones” in the ureters and cloaca, can occur in birds. Ureteroliths and cloacoliths are firm, rough-surfaced aggregations of urates. They are uncommon, and the pathogenesis is unclear.

3. HISTORY AND PHYSICAL EXAMINATION

A complete history and thorough physical examination should be performed. The clinical signs are often non-specific and include lethargy, weakness, anorexia, polydipsia/polyuria, normal to low body condition, unilateral or bilateral leg paresis, coelomic mass or ascites, and subcutaneous or intraarticular tophi.

4. DIAGNOSTIC TESTS

Complete blood count, plasma biochemistry panel and survey radiography are the minimum data base required when diagnosing urinary disorders. Ancillary diagnostic tests include urinalysis, water deprivation tests, glomerular filtration rate tests, radiographs, ultrasound, intravenous excretory urography, renal scintigraphy, and renal endoscopy and biopsy. Additional tests to detect infectious disease agent (e.g. blood cultures, PCR) or antibodies, as well as toxin screen might be indicated based on the differential diagnoses.

Hematological changes, like normocytic-normochromic anemia, heterophilia, lymphopenia and monocytosis, have been reported in birds with different type of renal disease but area considered non-specific. Plasma uric acid can be useful as a screening tool to severe renal disease. It has been estimated that renal function must be below 30% of its original capacity before hyperuricemia develops. Uric acid is produced predominantly in the liver but also in smaller amounts in the kidney and pancreas, as the major end product of nitrogen metabolism in birds. In carnivorous birds, uric acid production is directly related to the amount of protein consumed and transient rises are normal following high-protein meals, so fasting samples are required in these species. Uric acid is cleared mainly via tubular secretion in the kidneys, which is largely independent of glomerular filtration, water resorption and urine flow rate. In severe dehydration or post-renal obstruction, the uric acid can be elevated. Plasma urea in birds is produced only in small amounts and is excreted by glomerular filtration. It is more significantly affected by the hydration status than uric acid and has been considered the most useful indicator of dehydration as a cause of kidney failure in birds. Plasma creatinine does not appear to be a useful indicator of renal disease in birds. Plasma electrolyte abnormalities can occur with renal disease in birds but are poorly studied. N-acetyl-b-D-glucosaminidase (NAG) is a renal tubular enzyme used as an early indicator of renal tubular damage. It can be measured on plasma or urine. Reference values with a small number of birds from experimental studies have been published in chickens,⁴ pigeons,⁵ grey parrot⁶ and Hispaniolan Amazon parrots.⁷

Urinalysis is not commonly performed in birds due to extensive post-renal modification of urine by mixing of urine with feces in cloaca and further reabsorption of water in cloaca and rectum. To obtain an uncontaminated, sterile urine sample, the ureters have to be catheterized in the urodeum. This is both technically challenging, and not routinely performed. Reference values for urinalysis have been reported in the pigeon,⁸ falcon⁹ and ostrich.¹⁰ The urine pH varies with the diet of the bird and is usually between 6.5 and 8. The normal specific gravity is low due to decreased numbers of mammalian nephrons with loops of Henle and lack of counter-current multiplier system. The normal urine is colorless but is easily modified post-renal. Bacteria are normally found (passage through the cloaca) as it is trace of protein and glucose. Urate crystals are expected in the sediment, but the presence of cast is always an abnormal finding indicative of tubular disease. Biliverdinuria is a common finding in birds with liver disease. Heavy metal toxicity in Amazon parrots causes hemoglobinuria, leading to a red or brown urate color. The presence of blood in urine can be caused by kidney disease, but also could be blood from cloacal, reproductive, or gastrointestinal origins.

Water deprivation test can help to rule out unknown causes of polyuria/polydipsia, including central and nephrogenic diabetes insipidus, and psychogenic polydipsia.¹¹ A water deprivation test is carefully performed using a simple cage. The bird is weighed, and blood and urine are collected. Evaluate the packed cell volume (PCV), total solids and osmolality of blood, and specific gravity and osmolality of urine. Place the avian patient in a cage with no food or water for the duration of the test. Evaluate both blood and urine parameters every 3 to 24 hours for 12 to 48 hours, depending on the species and physical condition of the bird. The bird's behavior and laboratory results give a presumptive diagnosis. Birds with psychogenic polydipsia should tolerate this test well and develop more concentrated urine (increased osmolality and specific gravity) and an increase in PCV, total solids and plasma osmolality, all consistent with dehydration.¹² Birds with central (lack of production of arginine vasotocin [AVT]) or nephrogenic (inadequate response to AVT) diabetes insipidus should have different results than those with psychogenic causes. Birds with diabetes insipidus become dehydrated (as supported by plasma variables) but maintain dilute urine (low specific gravity and osmolality).¹³

Glomerular filtration rate (GFR) can be used to quantitate renal function. Decrease in GFR has to be differentiated from a normal physiologic response to water restriction. Plasma exogenous excretion creatinine, iothexol and inulin has been used following IM or IV administration in a number of avian species to calculate plasma concentrations over time and the GFR.¹⁴⁻¹⁶

Survey radiographs are often used for initial assessment of the urinary tract. Sedation or general anesthesia facilitates exact positioning to yield the best image of the desired area. In the normal avian patient, the kidneys are often indistinct in the ventral dorsal view and the cranial end of the kidney is superimposed by the gonad in the lateral view. The kidneys do not extend ventrally to an imaginary horizontal line parallel to the spine that passes through the ventral border of the acetabulum.¹⁷ Renomegaly and mineralization of the kidneys can be observed in birds with renal disease. Gastrointestinal contrast studies can help to differentiate the kidneys from surrounding structures.

Ultrasonography is particularly helpful in cases of abdominal distension. Because of the dorsal position of the avian kidney and the presence of air sacs, ultrasound is generally impossible in the normal bird. Organomegaly, air sac compression, and ascites all facilitate transducer coupling and improve image quality. Patient safety is at risk when a patient with profound abdominal distension is placed in dorsal recumbency.

Intravenous excretory urography with iodinated contrast agents has been described in birds as a method to evaluate the urinary tract. Intravenous contrast studies are contraindicated in birds that are dehydrated or with severe renal insufficiency. Aorta, heart, and pulmonary arteries are demonstrated in seconds after injection. Kidneys and ureters are shown 30 to 60 seconds after injection. The ureters can be seen running parallel to the medial edge of the kidneys and ending in the cloaca. The cloaca and terminal gut are visualized 2 to 5 minutes after the injection.¹⁸ A reduction or complete absence of contrast flow through the kidney or ureter sometimes occurs when cysts, renal tumors, or ureteroliths are present.¹⁹

Endoscopy and biopsy allow direct visualization of the complete urinary system (kidneys, ureters and cloaca). Small yellow or white spots within the kidney are signs of urate stasis or gout. A filled and swollen ureter might be due to obstruction or dehydration. Loss of the typical surface structure of the kidney is often a sign of glomerulonephrosis while larger single yellow masses may be nephritis, abscesses or neoplasia. Renal biopsies are necessary to differentiate different conditions. Renal biopsies can be collected from the cranial, middle, or caudal divisions

of the kidney from within the abdominal air sacs. In general, there is no need to use scissors. Endoscopy can also be used to diagnose enterolith or cloacoliths.

Fine needle aspirate, cytology and murexide test are needed to confirm the presence of urate tophi. Macroscopically, the aspirated material from articular gout looks like toothpaste. The presence of urate can be confirmed by performing the murexide test or by microscopic examination of aspirates from suspected tophi. The murexide test is performed by mixing a drop of nitric acid with a small amount of the suspected material on a slide. The material is dried by evaporation in a flame and allowed to cool. One drop of concentrated ammonia is added, and if urates are present, a pale purple color will develop. Microscopically, sharp, needle-shaped crystals can be seen in smears.¹

5. TREATMENT

The treatment of the different disorders of the urinary system require addressing the primary cause targeting the therapy to a specific etiology, while providing supportive care to recover from the primary lesions and preventing secondary damage.

Fluid therapy constitutes one of the most important treatments in cases of kidney disorders in birds. Uric acid is eliminated by active tubular secretion and water is needed to eliminate the suspension through the renal tubules. Without regular removal by diuresis, urates can accumulate within the kidneys. Fluid type is selected based on results of biochemical analyses, evaluation of blood electrolytes, glucose, and acid–base status. When these values are not known, a balanced isotonic crystalloid solution, such as lactated Ringer's solution may be used for rehydration and hemodynamic support. Caution is recommended when using colloid fluids in patients with renal disease by extrapolation from mammals

Dietary modification using commercial diets low in proteins formulated for birds with renal insufficiency (eg, Roudybush AK formula; Woodland, CA) has been suggested, although evidence-based data on whether protein restriction is beneficial in birds are lacking. Renal lesions, such as gout, have been associated with excess dietary protein in birds, but only under specific conditions. A safe recommendation is that birds with hyperuricemia should not consume diets with protein levels greater than what is

Omega-3 fatty acid and vitamin A supplementation have been suggested as supportive treatment, but only anecdotal information exists regarding their use in renal disease, and it is extrapolated from small animal medicine.

Antibiotics are indicated in cases of confirmed or suspected bacterial nephritis for at least 4-6 weeks. Ideally, antibiotics would be selected based on culture and susceptibility. If this is not possible, broad-spectrum antibiotics are recommended. Nonsteroidal anti-inflammatories, while contraindicated with compromised renal function, has been suggested at low doses to treat inflammation in the kidneys but no studies evaluating these potential benefits have been published.

Xanthine oxidase inhibitors, such as allopurinol, decrease uric acid synthesis. The use of allopurinol in avian medicine is controversial since uric acid is a potent antioxidant and less toxic than its precursors, and information with controlled studies is available for only a limited number of species. In broilers, uricemia was reduced in birds treated with allopurinol (25-50 mg/kg by mouth), but reduced body mass in growing chicks.^{20,21} In red-tailed hawks (*Buteo jamaicensis*) with induced hyperuricemia, allopurinol has been shown to be toxic at 50 mg/kg by mouth every 24 hours and was attributed to the accumulation of oxypurinol, a metabolite that worsens renal gout.²² Allopurinol given at 25 mg/kg by mouth every 24 hours to red-tailed hawks was shown

to be safe but had no significant effect on plasma uric acid concentrations. Based on the lack of response at a dose of 25 mg/kg and the toxic effects at 50 mg/kg, allopurinol is not recommended in the red-tailed hawk.²³ It is unknown whether this finding should be extrapolated to psittacine birds.

Uricase oxidizes uric acid to allantoin in humans. Poultry on a high-protein diet developed hyperuricemia, which was reversed with uricase injection.³ In pigeons (*Columba livia domestica*) and in red-tailed hawks, the uricolytic properties of uricase have also been studied. Plasma concentrations of allantoin and uric acid were evaluated before and after receiving 100, 200, and 600 UI/kg uricase intramuscularly once daily. All treatments caused a significant decrease in plasma uric acid concentrations within 2 days after the first administration, when compared with controls. Plasma allantoin concentrations were also significantly higher when compared with controls, suggesting a similar mechanism of action in these species.²⁴ In healthy grey parrots, 0.5 mg/kg IM decreased uric acid markedly 16-24 hours after administration.

The xanthine dehydrogenase inhibitor, colchicine, is used for its antigout activity in humans. Colchicine has also been used to treat amyloidosis and renal fibrosis in small animals. In turkeys diagnosed with articular gout, colchicine administered at 0.18 mg/kg by mouth every 24 hours for 7 days failed to influence uric acid concentrations.²⁵ No controlled study on colchicine has been published in birds.

Uricosuric drugs, such as probenecid, promote uric acid excretion by the kidneys in mammals mostly by reducing the absorption of uric acid in the tubuli. Birds lack the reabsorptive mechanism of uric acid and probenecid could further reduce the active secretion of uric acid into the tubuli. Uricosuric drugs are also contraindicated when tubular urate crystals are present, which is frequently seen in birds.²⁶ In chickens, probenecid at 100 mg/kg IV has shown to diminish renal tubular secretion of uric acid and to raise markedly plasma urate concentrations.²⁷ There are no controlled studies evaluating the efficacy and safety of probenecid in birds with hyperuricemia.

In the case of uroliths and cloacoliths, surgery might be needed to remove the stone.^{19,28}

TABLES

Table 1. Urinalysis values reported in birds

Parameter	Reference Range	Pigeon	Falcon	Ostrich
Specific gravity (g/mL)	1.005-1.020	—	—	—
pH	6.4-8 Laying hens and carnivorous birds may have more acidic urine; cloacal contents may alter urine pH	5.5-6.9	5-7	7.6 ± 1.5
Protein (g/dL)	Negative to trace	0.11-1.99	0.3 ± 0.2	2.6 ± 1.5

Glucose (mg/dL)	Negative to trace	—	24.3 ± 39.6	Negative
Ketones	Negative; ketonuria is sometimes present in migratory birds	—	Negative	Negative
Bilirubin	Negative	—	—	Negative
Urobilino gen	Negative	—	—	Negative

a 95% confident interval.

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